

Topics in Analysis (3TN) Information Sheet - 2024

Welcome to the 3TN team!

Advanced courses in Mathematical Analysis, require that students have mastered topics and concepts of Real Analysis such as the concepts of supremum, infimum, maximum, minimum and the study of sequences and series of real and complex numbers, sequences of functions, and topological concepts such as metric spaces.

To make sure that you have a proper knowledge of those concepts, we will start this module with a thorough revision of some of those topics.

We intend to create an environment where proper learning, collaboration, cooperation, and mutual respect are effective. We will run the module face-to-face with lectures in the venue to be recorded. However, we are also committed to move lectures and tutorials online if it is necessary. For instance, when there are interruptions due to protests and other incidents which require that we vacate campus.

As you have seen from your previous courses like MAM2015S and 3MS in MAM3000W, the study of topics in Analysis can be highly demanding. Therefore, it requires a good planning and organization that will give space to continuous study throughout the semester.

The module Topics in Analysis will be run as follows:

1. All the written notes will be available. I will announce the written notes and the lecture videos of the week on the **3TN lessons tab** together with a tutorial sheet of that week.
2. According to the timetable, 3TN lectures will take place at 12h00 (5th period) in **M209** on Mondays, Thursdays, and the following Wednesdays: **31 Jul; 14, 28 Aug; 18 Sep; 9 Oct.**
3. According to the timetable, there will be a tutorial session on Thursdays 13h00-15h00, starting from next week (22-26 July). The tutorial sheets will be uploaded on Vula on Mondays, giving you enough time to work through before the tutorial session.
4. Consultations: I am available for consultations provided an email was sent to arrange the meeting date and time. My office number is **314** (3rd floor) in the Maths building.
5. Assessment: There will be **two class tests** to be written on the following dates and times:

Class test 1 (CT1): Monday 26 August (18h00-20h00) in LS 1A and LS 1B.

Class test 2 (CT2): Monday 7 October (18h00-20h00) in M304.

Before each class test, there will a mini-test to be written in the tutorial session (14h00-15h00) and that will help you check on the knowledge of the course material and your readiness to write the coming class test.

The mini-test dates: Mini test1 (MT1): Thursday 15 August; Mini-test2 (MT2): Thursday 19 September.

The class record CR for 3TN is calculated as follows: **$CR = 0.2 \times AMT + 0.8 \times ACT$** , where **AMT is the average of the mini-tests and ACT is the average of the class tests.**

The final mark (FM) for this module will be calculated as

$$FM = \max(0.4 \times CR + 0.6 \times EX, 0.2 \times CR + 0.8 \times EX).$$

DP requirement: $CR \geq 30\%$.

Topics in Analysis (3TN) – Course content 2024

1. Sequences

- Basic definitions on sequences of numbers
- Study of the behaviour of sequences of real numbers

2. Revision on metric spaces

- Distance, balls, open and closed sets, limit of a function, convergence of sequences, compactness, completeness, etc.
- Spaces of continuous functions between metric spaces (continuity, uniform continuity, Lipschitz continuity, Hölderian continuity, etc.
- Pointwise convergence and uniform convergence for sequences of functions.

3. Normed spaces and linear continuous mappings between normed spaces

- Norms (equivalence of norms), Banach spaces
- Linear continuous mappings between normed spaces (norm of a linear continuous mapping, the dual space).

4. Hilbert spaces

- Scalar product (parallelogram identity) - Hilbert spaces
- The orthogonal complement, the orthogonal projection operator, the Gram-Schmidt orthonormalization process.

5. Differential calculus between normed spaces

- Gâteaux and Fréchet differentiability
- Chain rule
- Lagrange inequality
- Convexity (convex Fréchet differentiable functions)
- Implicit function theorem (IFT) – Inverse mapping theorem (IMT)

6. On Riemann integration and its limitations (time permitting)